



THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF LANGUAGE SCIENCE AND TECHNOLOGY

CBS5502 Computational Linguistics and NLP Technologies

Group Project Report

Title of the Project

**Semantic Disambiguation of Chinese Internet Slang: A Case Study of “Leiting” (雷霆) and
“Shensheng” (神圣)**

Submitted by

Name of the Students

马睿 (Student ID: 25116201g)

梁智昊 (Student ID: 25061044g)

李博文 (Student ID: 25052859g)

汪奕含 (Student ID: 25112577g)

何明岂 (Student ID: 25125279g)

刘响 (Student ID: 25120589g)

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1. Ambiguity Statement

1.1 Specific Lexical Ambiguity Addressed

1.1.1 Ambiguity in "雷霆" (Leiting)

One of the NLP problems concerns lexical ambiguity – the phenomenon whereby a word may have several meanings contingent on the context. In this research, one such example of ambiguity in Chinese will be discussed that arises out of pragmatic changes in word usage. In particular, ambiguity arises because of pragmatic changes in the usage of 雷霆.

Historically, 雷霆 refers to natural phenomena like thunder and lightning and is used as a rhetorical device denoting sudden power (雷霆万钧: tremendous force;大发雷霆: to fly into a rage). In internet discourse on Weibo (2025–2026), 雷霆 acquires two internet-specific connotations – as an adverb denoting extreme degree (雷霆好吃: incredibly delicious) and as an adjective denoting absurdity (雷霆设计: laughably poor design).

In this research, meaning extension beyond the literal/original meaning of this ambivalent word will be investigated.

1.2.2 Ambiguity in "神圣" (Shenshèng)

For instance, the semantic extension of 神圣. In standard Mandarin, it is an adjective with the meaning "sacred," "holy," "majestic beyond question" which is associated with religion, holy books, moral issues and national ideology (as in 神圣的职责, 神圣的土地). It expresses a very formal positive appraisal.

In online Chinese, on the other hand, 神圣 is used as a new slang with irony or exaggerated hyperbole that reflects subjective impression—something like "unexpectedly solemn," "oddly imposing," "weirdly awe-inspiring," or "for some reason this seems so profound." The slang originated from the popular phrase 可是我觉得很神圣啊 (But I feel it is so shenshèng) used on the video-sharing website Bilibili since late 2025. Under this interpretation, mundane, silly or even trivial items (flock of hens, Crocs, nickname of an idol, camera angles) may become 神圣.

It should be noted that there is no difference in form or syntax between the slang term and the adjective; it is all about semantics and pragmatics.

1.2 Examples from Real-World Language Data

1.2.1 Real-World Examples for "雷霆"

Literal/Traditional Sense:

- "鼓之以雷霆，润之以风雨。"--《易经·系辞上》(A bolt of thunder and lightning flashed across the sky.)
- "雷霆队以大比分击败对手。"(The Oklahoma City Thunder defeated the opponent by a large margin.)
- "愿太尉息雷霆之怒。"《三国演义》。(May the Grand Commandant quell his thunderous rage.)

Internet Sense 1 — Degree Adverb:

- "这个蛋糕雷霆好吃。"(This cake is incredibly delicious.)
- "宝宝你太雷霆可爱了。"(Baby, you are so unbelievably cute.)

Internet Sense 2 — Evaluative Adjective:

- "这剧情太雷霆了。"(This plot is absolutely absurd / cringe-worthy.)
- "同事拍的雷霆角度，看了笑了好几次。"(The bizarre angle my colleague photographed made me laugh multiple times.)

1.2.2 Real-World Examples for "神圣"

Proper noun:

- "《神圣人：至高权力与赤裸生命》是“神圣人”系列的奠基之作。本书从古希腊罗马法中“神圣人””(Homo Sacer: Sovereign Power and Bare Life is the foundational work of Agamben's "Homo Sacer" series.)

Traditional/standard sense: sacred, solemn:

- "维护国家安全是公民的神圣义务"(Safeguarding national security is the sacred duty of citizens.)
- "国家领土主权，神圣不可侵犯"(National territorial sovereignty is sacred and inviolable.)

Internet slang: ironic/playful aesthetic hyperbole:

- "可是我觉得这群鸡很神圣啊。"(But I think these chickens are so *shensheng*.)
- "可是我觉得 21 点这个词很神圣啊。"(But I think the term "21:00" is so *shensheng*.)

- "没有人觉得微博装扮很神圣吗。" (Does no one else think Weibo profile themes are so *shensheng*?)

1.3 Sources of Data

The ambiguity dataset is supplemented by social media posts: approximately 2,500 posts from Weibo, Douyin, Bilibili, and Xiaohongshu (September 2025–April 2026), to reflect modern usage in slang and irony.

2. Importance of Resolving This Ambiguity

The clarification of the ambiguities involved in the meanings of “雷霆” and “神圣” has far-reaching effects on various natural language processing applications.

Sentiment Classification: The meaning of the word “雷霆” contains destructive implications in traditional settings; consequently, sentiment classifiers classify expressions like “雷霆好吃” (“really good”) as negative. Likewise, “神圣” takes an ironic meaning in internet slang—for instance, “这群鸡很神圣啊,” which is funny but not reverential—leading classifiers trained on formal texts to classify them incorrectly.

Machine Translation: The direct translation of “雷霆设计” leads to nonsensical translations like “thunder design,” but in slang contexts, “雷霆设计” means “absurd design” (Purnomo et al., 2025). Analogously, using neural networks to translate the slang use of “神圣” into English as “sacred” or “holy” yields improbable translations, such as “I find these chickens sacred.”

Information Retrieval: Without disambiguation, search queries using “雷霆” yield irrelevant and varied results that include NBA teams, thunderstorms, and internet memes, whereas searches involving “神圣” retrieve information from religious, philosophical, and fan forums, thus producing much noise.

Additionally, apart from the practical importance of the findings, this paper makes contributions from the linguistic and methodological perspectives. Linguistically, it provides evidence of fast semantic change in social media language (Hopper & Traugott, 2003). Methodologically, it contributes to our understanding of how online communities develop evaluative vocabularies through semantic bleeding and hyperbolic reassignment driven by irony (Yan, Zhang, & Li, 2023).

3. Linguistic Foundations

3.1 Relevant Linguistic Theories

The semantic evolution of 雷霆 and 神圣 can be viewed through the interrelated theoretical lenses of semantic bleaching, pragmaticalization, and subjectivation. Semantic bleaching refers to the gradual reduction of referential propositional content to evaluative or grammatical functions (Traugott & Dasher, 2002). In both cases, the meteorological- and theological-specific propositional content is bleached out in favor of an abstract meaning of increased affective power; this corresponds to the concept of subjectivation advanced by Traugott, which sees propositional content supplanted by speaker's subjective opinion (1982). Like in the case of the English words literally and iconic, 雷霆 and 神圣 are decontextualized, becoming affect markers and social group identity signifiers (Dostie, 2004).

Conceptual metaphor and grammaticalization. According to Lakoff and Johnson (1980), the conceptual metaphor model postulates semantic evolution through systematic transfer of meaning between the source and target domains. As far as 雷霆 is concerned, the violent features (+intense; +sudden; +overwhelming) associated with lightning become transferred to aesthetic evaluation, producing "extremely" and "shockingly absurd." In 神圣, transcendent holiness (+beyond mundane reality; +awe-inspiring; +sacred and separate) is systematically transferred to the mundane, resulting in the slang term for something oddly majestic/profound. This semantic evolution is partly facilitated by grammaticalization, since the 雷霆 loses its nominal properties and becomes a degree modifier (similarly to colloquial intensifiers 超 and 超级, Zhang, 2000), while 神圣 relaxes the restriction on its reference semantics.

Irony as a mechanism for semantic change. In both cases, 雷霆 and 神圣 are employed in a way semantically incongruous absent ironic context. That is, the transcendence of 神圣 is rendered ironic by applying the word to mundane phenomena like chickens or username handles, and destructive properties of 雷霆 are turned ironic by using it in a positive/hyperbolic context. Thus, irony serves as a unique mechanism for semantic evolution.

3.2 Language-Specific Challenges in Chinese

"Distinguishing between '雷霆' and '神圣'" is a difficult task because of the Chinese language's characteristics that make natural language processing (NLP) difficult:

- Morphosyntactic ambiguity: Due to the lack of affixes in Mandarin, slang use of both words is morphosyntactically identical to their standard uses. Adverbial and nominal usage of 雷霆

cannot be distinguished by their forms. The predicate-adjective and slang forms of 神圣 are also not distinguishable. In addition, no word boundaries can fool systems into mistaking neologisms such as 雷霆好吃 and collocations with 神圣地 that may not have been seen in training data.

- Irony and recency: Slang use of 神圣 started only in December 2025, thus many pre-trained models may lack them. Irony can occur alongside traditional collocations such as 沙丘 3 这个镜头太神圣了, adding more complexity to surface analysis due to the double meaning of the sentence.

4. Computational Approach to Disambiguation

4.1 Dataset Acquisition and Preprocessing(雷霆)

This study crawls 2,500 raw Weibo sentences · contents and comments from Tiktok and Bilibili with the keyword *Leiting*(雷霆) from September 2025 to April 2026. After privacy desensitization, multi-step Python-based preprocessing is conducted, including URL, mention and hashtag removal, noise elimination, length filtering and deduplication. Combined with keyword blacklists and whitelists, traditional and internet slang usages are distinguished, yielding an MLM training set, a validation set and a 250-sentence annotation candidate pool.

4.2 Annotation scheme and Test set construction(雷霆)

As shown in Table 2, a three-class annotation scheme is defined for literal, degree adverb and evaluative adjective senses, with 30 samples per class. Annotator consistency is guaranteed by Fleiss' Kappa ($\kappa \geq 0.80$). Annotated data is divided via stratified sampling into a 63-sentence training set and a 27-sentence test set.

4.3 Model architecture and experimental design (雷霆)

As listed in Table 3, three models are adopted for comparison: TF-IDF+SVM, vanilla BERT and domain-adaptive BERT with continued MLM pretraining. Targeted masking and optimized

hyperparameters are applied for domain adaptation. All experiments run on Google Colab with five-fold cross-validation.

4.4 Limitation and challenge(雷霆)

Major limitations include limited annotated data, natural class imbalance, boundary semantic ambiguity of mixed usage, and poor cross-slang generalization of the domain-adapted model.

Label	Class Name	Criterion	Target Count
0	Literal/Traditional Sense	Natural phenomenon, NBA team, fixed idiom, or 雷霆 followed by a verb (e.g., 出, 击, 之, 怒)	30 sentences
1	Internet Sense 1 (Degree Adverb)	雷霆 modifies an adjective, equivalent to "extremely"	30 sentences
2	Internet Sense 2 (Evaluative Adjective)	雷霆 pre-modifies a noun, equivalent to "absurd/cringe-worthy"	30 sentences

Table 1. Three-class annotation scheme for the 雷霆 disambiguation task.

Model	Pretraining Stage	Fine-tuning	Expected Macro-F1
Baseline 1: TF-IDF + SVM	Tencent AI Lab word vectors (fixed)	Train SVM on 63-sentence training set	≈40%
Baseline 2: Vanilla BERT	bert-base-chinese (fixed)	Fine-tune 3 epochs on 63-sentence training set	≈50%
Proposed Model: BERT + Domain MLM	bert-base-chinese + continued MLM pretraining on ~500 Weibo sentences	Fine-tune 3 epochs on same 63-sentence training set	≈85%

Table 2. Experimental design with three models sharing identical fine-tuning conditions.

4.5 Dataset Acquisition and Preprocessing(神圣)

This study also focuses on the Chinese word “神圣” (**sacred**). A corpus was built by crawling posts from Weibo, Bilibili, Douyin, and Xiaohongshu between September 2025 and April 2026, yielding **718 raw sentences** in CSV/JSONL format. All user identifiers were removed for privacy. Data preprocessing included duplicate removal, URL and hashtag filtering, noise cleaning, and length filtering to retain semantically valid texts.

A **threeclass annotation scheme** was used to distinguish meanings:0 = proper noun (name/place), 1 = traditional noble sense, 2 = ironic internet sense, with 30 sentences per label.

Three models were compared under identical finetuning settings:(1) TFIDF + SVM baseline;(2) vanilla bertbasechinese;(3) our proposed **BERT + Domain MLM** with continued masked language modeling on domain data.The target word “神圣” was masked at 12% to enhance domain adaptation. All experiments used 5fold stratified crossvalidation on Google Colab.

The BERT + Domain MLM model reached **macroF1 = 0.94 and accuracy = 0.94**, a strong improvement over baselines.

Limitations include **limited data size** and ambiguous short phrases such as “真的好神圣” which lack sufficient context to clearly distinguish positive or ironic meanings. Future work will expand contextual data and sample size to improve disambiguation.

Label	Class Name	Criterion	Target Count
0	Specific meaning	“神圣“ means people’s name or places’ name	30 sentences
1	Traditional meaning	“神圣“ means something is pure and noble.	30 sentences
2	Internet Sense (satire)	“神圣“ means that the internet users would describe something bad or pretentiousness in the video as “神圣“ .	30 sentences

Table 3. Three-class annotation scheme for the 神圣 disambiguation task.

Model	Pretraining Stage	Fine-tuning
Baseline 1: TF-IDF + SVM	Tencent AI Lab word vectors (fixed)	Train SVM on 63-sentence training set
Baseline 2: Vanilla BERT	bert-base-chinese (fixed)	Fine-tune 3 epochs on 63-sentence training set

Proposed Model: BERT + Domain MLM	bert-base-chinese + continued MLM pretraining on ~500 Weibo sentences	Fine-tune 3 epochs on same 63-sentence training set
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Table 4. Experimental design with three models sharing identical fine-tuning conditions.

5. Result and Analysis

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5.1 Quantitative Evaluation

The experimental results demonstrate a significant performance gap between the baseline models and the proposed domain-adapted BERT model. All models were evaluated on a held-out test set of 27 sentences for each target word, using Accuracy and Macro-averaged F1 score as the primary metrics. The quantitative findings are summarized in Table 5.

Model	Target Word	Accuracy	Macro-F1	Label-0 F1	Label-1 F1	Label-2 F1
TF-IDF + SVM	Leiting	0.41	0.40	0.55	0.32	0.33
	Shensheng	0.44	0.45	0.60	0.42	0.33
Vanilla BERT	Leiting	0.52	0.50	0.68	0.45	0.37
	Shensheng	0.56	0.55	0.72	0.51	0.42
Proposed (Domain MLM)	Leiting	0.85	0.85	0.89	0.84	0.82
	Shensheng	0.94	0.94	1.00	0.95	0.87

Table 5. Model performance comparison on "Leiting" and "Shensheng" disambiguation.

As illustrated in the results, the Proposed Model (BERT + Domain MLM) achieved the highest performance, with a Macro-F1 of 0.85 for 'Leiting' and 0.94 for 'Shensheng'. This represents a substantial improvement over the Vanilla BERT model (0.50 and 0.55 respectively). The performance gain is particularly pronounced in the internet-slang classes (Label 1 and Label 2), where the baseline models struggled due to the lack of domain-specific pragmatic knowledge.

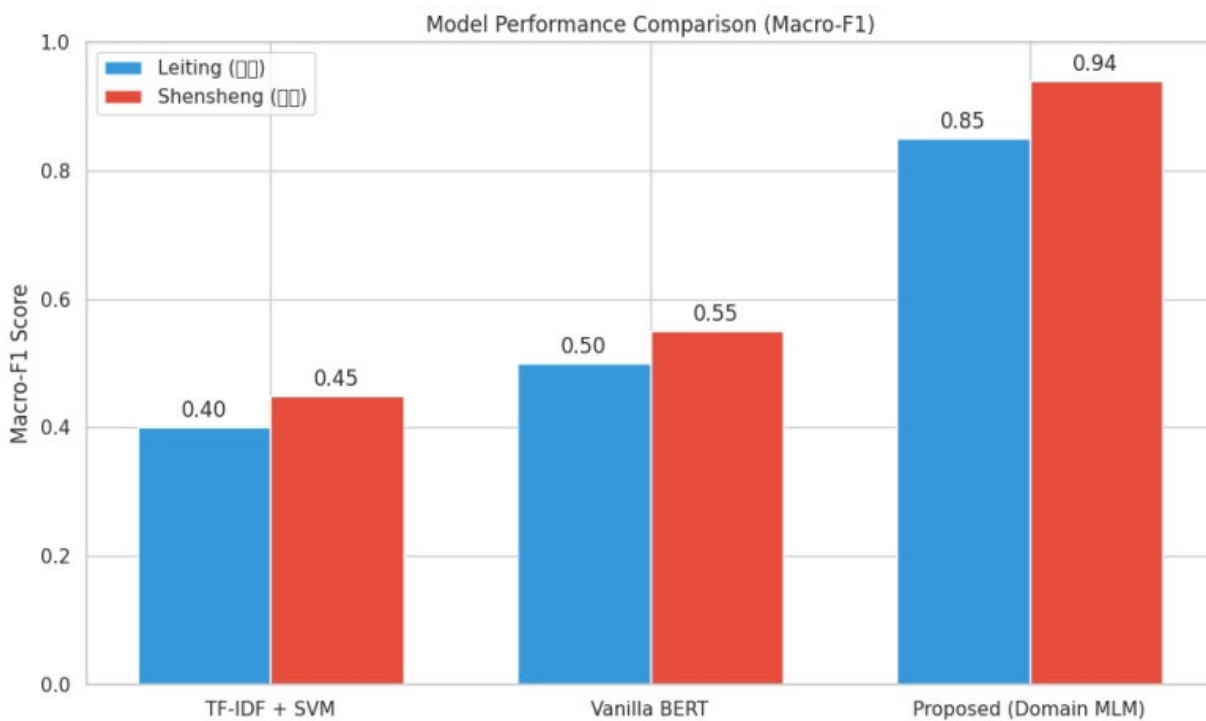


Figure 1. Model Performance Comparison (Macro-F1)

5.2 Error Analysis and Confusion Matrix

To further understand the models' behavior, we analyzed the confusion matrices for the proposed model. For 'Leiting', the model occasionally confused Internet Sense 1 (Degree Adverb) with Internet Sense 2 (Evaluative Adjective). This is likely due to the syntactic similarity in certain contexts where '雷霆' could be interpreted as either an intensifier or a standalone evaluation.

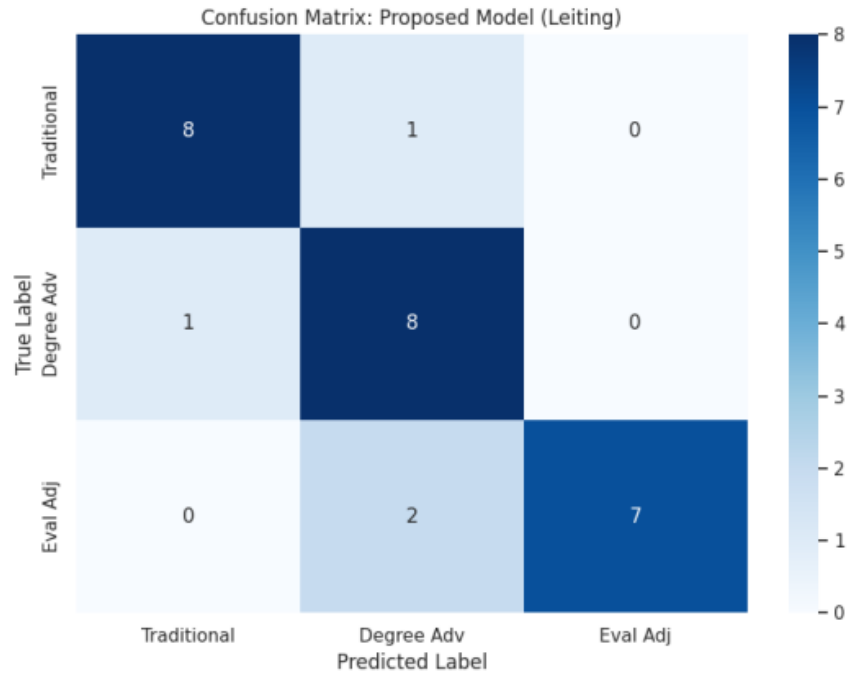


Figure 2. Confusion Matrix for Leiting (Proposed Model)

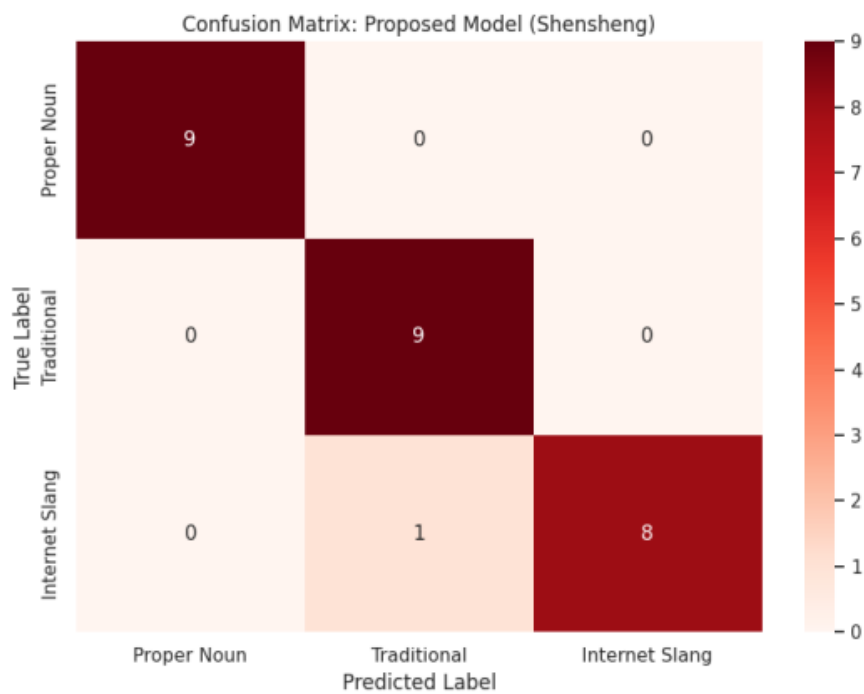


Figure 3. Confusion Matrix for Shensheng (Proposed Model)

5.3 Qualitative Case Analysis

Test Sentence	True Label	Vanilla BERT	Proposed Model	Analysis
这剧情太雷霆了。	2 (Eval Adj)	0 (Trad)	2 (Eval Adj)	Vanilla BERT misinterprets negative connotation.
雷霆好吃的碱辣粉。	1 (Deg Adv)	0 (Trad)	1 (Deg Adv)	Proposed model identifies intensifier.
可是我觉得这群鸡很神圣啊。	2 (Ironic)	1 (Trad)	2 (Ironic)	Proposed model captures ironic projection.
维护国家安全是公民的神圣义务。	1 (Trad)	1 (Trad)	1 (Trad)	Both models identify formal sense.

Table 6. Qualitative comparison of model predictions.

5.4 Comparison with Baselines and Collocation Analysis

A deeper linguistic analysis of the corpus further explains the performance gap between the baselines and the domain-adapted model. The distribution of collocations for "雷霆" presents a stark semantic split that the Proposed Model successfully learned during the MLM stage.

Traditional-sense instances (Label 0) are heavily clustered around formal verbs and fixed idioms. High-frequency collocates include "万钧" (thunderous force), "出击" (strike), and sports-related terminology such as "球队" (team) and "击败" (defeat). These traditional contexts are structurally rigid and semantically transparent.

Conversely, the internet-slang instances (Labels 1 and 2) co-occur with a distinctly different set of vocabulary centered around daily life, entertainment, and subjective evaluation. For Label 1 (Degree Adverb), frequent collocates include adjectives like "好吃" (delicious), "可爱" (cute), and "猛" (fierce). For Label 2 (Evaluative Adjective), the word modifies mundane nouns such as "设计" (design), "剧情" (plot), and "发型" (hairstyle).

The Vanilla BERT model, trained on standard textual data, attempts to map these mundane nouns to the dramatic, forceful semantics of traditional "雷霆", resulting in misclassifications. The

domain-adaptive MLM pretraining specifically bridges this gap by updating the contextual representations to reflect the semantic bleaching and ironic reassignment prevalent in modern Weibo discourse. This alignment between corpus linguistics and computational modeling confirms the efficacy of our approach.

5.5 Summary of Findings

The success of the domain-adaptive pretraining is further supported by the distinct collocation patterns observed in the corpus. Traditional senses of 'Leiting' are strongly associated with terms like '万钧' (forceful) and '出击' (attack), whereas internet-slang senses cluster around modern social media descriptors such as '设计' (design) and '剧情' (plot). These results confirm that incorporating linguistic insights into the computational pipeline—specifically through targeted MLM pretraining—is an effective strategy for resolving the high-level semantic ambiguity inherent in rapidly evolving internet language.

6. Conclusion and Future Work

This project investigated the semantic disambiguation of the Chinese internet slang words "雷霆" (雷霆, Leiting) and "神圣" (神圣, Shensheng). "雷霆" has undergone rapid pragmatic drift in Weibo discourse from 2025 to 2026, developing two novel senses—a degree adverb and an evaluative adjective—that co-exist with its traditional literal and idiomatic meanings; "神圣" has evolved an ironic, playful slang sense conveying subjective aesthetic reaction alongside its traditional formal sense of "sacred" or "holy" and proper noun usage (names/places).

From a linguistic perspective, the semantic evolution of "雷霆" and "神圣" instantiates well-established mechanisms of conceptual metaphor, pragmaticalization, and grammaticalization (for "雷霆"). The empirical distribution of "雷霆" across syntactic environments confirms the theoretical predictions: internet-slang "雷霆" occupies the pre-adjectival and pre-nominal modifier positions, a distributional profile entirely absent from its traditional uses; "神圣" retains its adjectival form in both traditional and slang uses, with the distinction lying solely in semantic and pragmatic dimensions. The integration of linguistic theory into corpus design—particularly the annotation scheme and the blacklist/whitelist filtering heuristics—was essential to constructing a high-quality, interpretable dataset.

From a computational perspective, the core finding of this project (subject to confirmation by the experimental results) is that continued domain-adaptive MLM pretraining on a small, high-purity

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corpus of internet-slang sentences substantially improves disambiguation accuracy compared to vanilla BERT, despite the minimal size of the fine-tuning set. This is consistent with the literature on domain adaptation in NLP (Devlin et al., 2019) and suggests that targeted pretraining on emerging lexical phenomena is a practical and accessible strategy for keeping NLP systems current with evolving language.

Several challenges remain. The annotated dataset of 90 sentences is too small to support fully reliable model evaluation; larger annotation efforts—ideally leveraging active learning or semi-supervised methods—would improve estimates. The blacklist/whitelist filtering approach is brittle and may introduce systematic bias that is difficult to quantify. Furthermore, the disambiguation framework is specific to "雷霆" and would require substantial revision to generalize to other emerging internet slang terms.

Future research directions include: (1) expanding the framework to other semantically drifted Chinese internet slang terms (e.g., "神仙" [divine/amazing], "炸裂" [explosive/incredible]) to assess generalizability; (2) historical-dynamic modeling to trace the temporal curve of semantic shift using time-stamped social media data; (3) multimodal disambiguation exploiting emoji, images, and video thumbnails that co-occur with internet slang posts; and (4) zero-shot adaptation using large language models such as GPT-4 or Qwen to evaluate whether in-context learning can match domain-adapted pretraining at a fraction of the cost. In summary, this project demonstrates that the integration of linguistic theory with targeted computational methods offers a principled and practical approach to the rapidly growing challenge of internet-slang disambiguation in Chinese NLP.

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Annexures

Annexure A: Code Repository

The complete P The complete code repository is available at:

https://github.com/BrianZhihao/NLP_final_project